ASE CHAPTER 4

4 Academic Vocabulary & Syllabus Ledger · Software Project Planning

e 1: Software Size Estimation

Lines of Code (LOC)

mic Definition & Technical Specification: Lines of Code (LOC) is a code-based measure of software size, ssed as a count of source-code lines under a stated counting convention.

Mechanisms & Standards:

1. Counting Conventions:

Physical-line and logical-statement counts can differ. A reported count should identify its convention; treatment of blank lines, comments, and language syntax may vary.

2. Critical Engineering Limitations:

Language Dependence: Expressing the exact same algorithm requires significantly fewer lines in high-level languages (e.g., Python, SQL) than in low-level languages (e.g., Assembly, C).

LOC is language- and implementation-dependent, so a larger count alone does not establish greater user functionality or better productivity.

: K.K. Aggarwal & Yogesh Singh (3rd ed.), §4.3.1, pp. 143–145; Companion Slides 6–8.

. Function Count and Its Relation to Function Point Analysis

mic Definition & Technical Specification: The course lists Function Count as a size-estimation topic. The chapter ns Albrecht's Function Point method: it classifies five types of functional units by complexity and weights them culate unadjusted and adjusted Function Points. Keep the course term Function Count distinct from the ated measures UFP and FP.

ematical Formulation & Pipeline:

1. The 5 Information Domain Characteristics:

External Inputs (EI): Transactions entering the system that alter internal state (Weights: Low=3, Average=4, High=6).

External Outputs (EO): Data or information leaving the system as an output (Weights: Low=4, Average=5, High=7).

External Inquiries (EQ): On-demand searches that display data without altering internal state (Weights: Low=3, Average=4, High=6).

Internal Logical Files (ILF): User-identifiable logical data groups maintained entirely inside the application boundary (Weights: Low=7, Average=10, High=15).

External Interface Files (EIF): Reference data files maintained by external systems across interfaces (Weights: Low=5, Average=7, High=10).

2. Unadjusted Function Points (UFP):

$$\text{UFP} = \sum_{i=1}^{5} \sum_{j=1}^{3} W_{ij} \cdot Z_{ij}$$

3. Complexity Adjustment Factor (CAF):

$$\text{CAF} = 0.65 + 0.01 \times \sum_{i=1}^{14} F_i \quad (0.65 \leq \text{CAF} \leq 1.35)$$

4. Final Adjusted Function Points:

$$\text{FP} = \text{UFP} \times \text{CAF}$$

: K.K. Aggarwal & Yogesh Singh (3rd ed.), §4.3.2, pp. 145–148; Companion Slides 9–30.

---

e 2: Cost Estimation Models

. Static, Single Variable Models

mic Definition & Technical Specification: Static Single-Variable Models are empirical cost estimation equations redict total development effort as a mathematical power function of a single project parameter dominantly software size in thousands of lines of code (KLOC).

ematical Formulation & Canonical Variants:

1. General Power Equation:

$$E = a \times L^{b} \, [\text{Person-Months}]$$

Where a is an organization-specific baseline constant, L is size in KLOC, and b is the scaling exponent.

2. IBM Walston-Felix Model (Federal Systems Division):

$$E = 5.2 \times L^{0.91}$$

3. NASA Software Engineering Laboratory (SEL) Model (Goddard Space Flight Center):

$$E = 1.4 \times L^{0.93}$$

: These equations use one project-size variable, L in KLOC, to estimate effort. Their coefficients and calibration
ts differ; do not infer which model is more accurate for a new project from the equations alone.

: K.K. Aggarwal & Yogesh Singh (3rd ed.), §4.3.3, pp. 149–151; Companion Slides 34–35, 37–40.

📝 **ملاحظة أكاديمية هامة (Academic Note)**

الشرح المفاهيمي وفخ الامتحان:

- **مفهوم المصطلح:** نماذج تجريبية تستخدم متغيراً وحيداً فقط (الحجم L) لتقدير الجهد.
- **نقطة تمييز:** للنموذجين معاملات وأسس مختلفة؛ اختلاف الناتج لا يثبت وحده أيهما أدق لمشروع جديد. قارن عند الحجم نفسه و
KLOC.

. Static, Multivariable Models

mic Definition & Technical Specification: Static multivariable models estimate software effort using size ner with other project or environment variables. The specific equation depends on the model.

ematical Formulation & Canonical Variants:

1. Basili and Freburger Multivariable Model:

$$E = a_0 + \sum_{i=1}^{n} a_i X_i$$

Effort is expressed as a multi-factor regression equation incorporating lines of code, program complexity, staffing experience, and memory constraints.

2. Walston-Felix productivity index:

$$I = \sum_{i=1}^{29} W_i X_i, \qquad X_i \in \{-1, 0, +1\}$$

The 29 assessed factors contribute weighted values to an index used with the source's productivity-index chart. Do not replace this procedure with a product of 29 multipliers.

ction from Single-Variable Models: - Multivariable approaches use project or environment factors in addition to size; culation depends on the specific model.

: K.K. Aggarwal & Yogesh Singh (3rd ed.), §4.3.4, pp. 151–152; Companion Slides 34, 36–40.

---

e 3: The Constructive Cost Model (COCOMO '81)

. Basic COCOMO Model

mic Definition & Technical Specification: Basic COCOMO is a size-based cost-estimation model that calculates and development time from KLOC using different coefficients for the Organic, Semidetached, and Embedded s.

ematical Formulations & Mode Equations:

1. The Three Operating Modes:

Organic Mode: Typically smaller projects with experienced teams, familiar domains, and relatively flexible constraints.

$$E = 2.4(\mathrm{KLOC})^{1.05}\ \mathrm{PM}, \quad D = 2.5(E)^{0.38}\ \mathrm{Months}$$

Semidetached Mode: Typically intermediate projects with mixed team experience and constraints.

$$E = 3.0(\mathrm{KLOC})^{1.12}\ \mathrm{PM}, \quad D = 2.5(E)^{0.35}\ \mathrm{Months}$$

Embedded Mode: Projects operating under tight hardware, software, or operational constraints.

$$E = 3.6(\mathrm{KLOC})^{1.20}\ \mathrm{PM}, \quad D = 2.5(E)^{0.32}\ \mathrm{Months}$$

: K.K. Aggarwal & Yogesh Singh (3rd ed.), §4.4.1, pp. 152–156; Companion Slides 41–50.

ملاحظة أكاديمية هامة (Academic Note)

الشرح المفاهيمي وفخ الامتحان:

- **مفهوم المصطلح:** المستوى التأسيسي الأول من COCOMO، يعتمد على الحجم ونمط بيئة المشروع فقط.
- **نقطة تمييز:** استعمل معاملات الحجم والجهد والوقت الخاصة بالنمط المختار؛ أوصاف الأنماط تساعد على الاختيار ولا تجعل حـ
  الحجم وحدها قاعدة مطلقة.

. Intermediate COCOMO Model

mic Definition & Technical Specification: Intermediate COCOMO extends the basic model by adjusting nominal using an Effort Adjustment Factor (EAF) derived from 15 cost-driver attributes spanning product, computer, nnel, and project dimensions.

ematical Formulation:

1. Governing Equations:

$$E = a_i(\text{KLOC})^{b_i} \times \text{EAF} \ [\text{Person-Months}]$$

$$\text{EAF} = \prod_{j=1}^{15} EM_j$$

2. Intermediate Leading Coefficients (a_i):

Organic: $a_i = 3.2, \quad b_i = 1.05$

Semidetached: $a_i = 3.0, \quad b_i = 1.12$

Embedded: $a_i = 2.8, \quad b_i = 1.20$

3. The 15 Cost Drivers (4 Categories):

Product Attributes: Reliability (RELY), Database Size (DATA), Product Complexity (CPLX).

Computer Attributes: Execution Time Constraint (TIME), Main Storage Constraint (STOR), Virtual Machine Volatility (VIRT), Turnaround Time (TURN).

Personnel Attributes: Analyst Capability (ACAP), Applications Experience (AEXP), Programmer Capability (PCAP), Virtual Machine Experience (VEXP), Programming Language Experience (LEXP).

Project Attributes: Modern Programming Practices (MODP), Use of Software Tools (TOOL), Required Development Schedule (SCED).

: K.K. Aggarwal & Yogesh Singh (3rd ed.), §4.4.2, pp. 156–158; Companion Slides 51–57.

ملاحظة أكاديمية هامة (Academic Note)

الشرح المفاهيمي وفخ الامتحان:

- **مفهوم المصطلح:** النموذج المتوسط الذي يدمج تأثير العوامل البيئية الـ 15 عبر ضرب معاملاتها للحصول على معامل التعديل
- **فخ الامتحان الأكبر:** المعامل الأساسي a_i في Intermediate يختلف عن Basic: في النمط العضوي (Organic) يرتفع من 2.4 إلى
  بينما في Semidetached يبقى 3.0، وفي Embedded ينخفض من 3.6 إلى 2.8. تذكر أن EAF ينتج عن ضرب جميع المعاملات م
  (وليس جمعها).

. Detailed COCOMO Model

mic Definition & Technical Specification: Detailed COCOMO incorporates all characteristics of the intermediate l while acknowledging that cost drivers impact individual development lifecycle phases unevenly, utilizing -sensitive effort multipliers and phase-specific effort/schedule distribution percentages.

Mechanisms & Lifecycle Phase Breakdown:

1. Phase-sensitive cost drivers: Detailed COCOMO applies cost-driver effects across five lifecycle phases:

$$\text{Plan \& Requirement} \rightarrow \text{System Design} \rightarrow \text{Detail Design} \rightarrow \text{Module Code \& Test} \rightarrow \text{Integration \& Test}$$

2. Three-Level Hierarchical Structure:
Module Level: Individual code units characterized by specific complexity and personnel assignments.
Subsystem Level: Aggregations of modules sharing common platform constraints.
System Level: Global project integration and overall management.
3. Curriculum Book Table 4.7 Phase Distribution (Aggarwal & Singh p. 159 / Slides 62–63):
Organic Small ($S \approx 2$): Plan & Requirement (6%), System Design (16%), Detail Design (26%), Module Code & Test (42%), Integration & Test (16%) for effort; the corresponding schedule fractions are 10%, 19%, 24%, 39%, and 18%.
(Audit Notice: Course source percentages sum to 106% for effort and 110% for schedule; preserved per course standard).

: K.K. Aggarwal & Yogesh Singh (3rd ed.), §4.4.3, pp. 158–161; Companion Slides 57–65.

📝 **ملاحظة أكاديمية هامة (Academic Note)**

الشرح المفاهيمي وفخ الامتحان:

- **مفهوم المصطلح:** يوزع النموذج أثر عوامل الكلفة وتقديرات الجهد والوقت على مراحل التطوير؛ استعمل أسماء المراحل كما يو الجدول.
- **نقطة تدقيق:** جدول Organic Small يعرض مجموع جهد 106% ومجموع وقت 110%؛ اذكر القيم المطبوعة كما هي عند الس عنها.

---

e 4: COCOMO II

. Application Composition Estimation Model

mic Definition & Technical Specification: The Application Composition model is a COCOMO II stage for ating applications assembled with reusable components, screens, reports, and 3GL modules.

ematical Formulation & Sizing Method:

1. Sizing Metric — Object Points (OP): Decomposes the system into Screens, Reports, and 3GL Modules.
2. Object Point Accounting:

$$\text{NOP} = \text{Object Points} \times \left(1 - \frac{\%\text{Reuse}}{100}\right)$$

3. Effort Computation via Productivity Rate (PROD):

$$\text{Effort} = \frac{\text{NOP}}{\text{PROD}} \ [\text{Person-Months}]$$

Where Developer Capability (PROD) ranges from Very Low ($4\ \text{NOP/PM}$), Nominal ($13\ \text{NOP/PM}$), to Very High ($50\ \text{NOP/PM}$).

: K.K. Aggarwal & Yogesh Singh (3rd ed.), §4.5.1, printed pp. 162–167; Companion Slides 73–84.

ملاحظة أكاديمية هامة (Academic Note)

الشرح المفاهيمي وفخ الامتحان:

- **مفهوم المصطلح:** المرحلة الأولى من COCOMO II، وتستخدم للمشاريع المبنية بأدوات التوليد السريع (مثل Visual Basic و e Forms) ونماذج الواجهات الأولية.
- **فخ الامتحان:** المقياس هنا هو **نقاط الكائنات (Object Points)** وليس Function Points ولا أسطر الكود. يتم احتساب الشاشات والتقارير ومكونات 3GL، ويتم خصم نسبة إعادة الاستخدام لحساب النتيجة الصافية (NOP).

. The Early Design Model

mic Definition & Technical Specification: The Early Design model is a COCOMO II stage used before a detailed ecture and its full cost-driver information are available. It estimates effort with seven aggregated cost drivers ve scale factors.

ematical Formulation:

1. Governing Equation:

$$E_{\text{nom}} = A \times (\text{Size})^{B} \times \prod_{i=1}^{7} EM_i$$

2. Sizing Metric: Size is expressed in KSLOC, typically derived by converting Unadjusted Function Points (UFP) using language gear ratios.

3. The 7 Aggregated Cost Drivers: Product Reliability and Complexity (RCPX), Required Reuse (RUSE), Platform Difficulty (PDIF), Personnel Capability (PERS), Personnel Experience (PREX), Facilities and Tools (FCIL), and Schedule (SCED).

: K.K. Aggarwal & Yogesh Singh (3rd ed.), §4.5.2, printed pp. 167–173; Companion Slides 85–89.

Post-Architecture Model

mic Definition & Technical Specification: The Post-Architecture Model is used when the software architecture is ble and project sizing and cost-driver information can be assessed in detail. It uses 17 effort multipliers and 5 factors.

ematical Formulation & Scale Factors:

1. Governing Effort Equation:

$$E = A \times (\text{Size})^B \times \prod_{i=1}^{17} EM_i \quad (A = 2.94)$$

2. Scaling Exponent (B) from 5 Scale Factors (SF_j):

$$B = 0.91 + 0.01 \times \sum_{j=1}^{5} SF_j$$

Precedentedness (PREC): Domain novelty (0.00 to 6.20).
Development Flexibility (FLEX): Constraint rigidity (0.00 to 5.07).
Architecture / Risk Resolution (RESL): Risk mitigation depth (0.00 to 7.07).
Team Cohesion (TEAM): Stakeholder synchronization (0.00 to 5.48).
Process Maturity (PMAT): CMM/CMMI rating (0.00 to 7.80).

: K.K. Aggarwal & Yogesh Singh (3rd ed.), §4.5.3, printed pp. 168–181; Companion Slides 90–120.

ملاحظة أكاديمية هامة (Academic Note)

الشرح المفاهيمي وفخ الامتحان:

- **مفهوم المصطلح:** يستخدم النموذج عندما تتوفر تفاصيل المعمارية ومدخلات المشروع بدرجة تسمح بتقدير أدق.
- **نقطة تمييز:** لا تخلط بين عوامل القياس الخمسة التي تحسب الأس B ومضاعفات الجهد السبعة عشر التي تعدّل الجهد.

---

e 5: The Putnam Resource Allocation Model

The Norden/Rayleigh Curve

mic Definition & Technical Specification: The Norden/Rayleigh curve models the rate of manpower use over t time; its peak occurs at the development-time parameter t_d in the chapter's formulation.

ematical Formulation & Peak Manning Formula:

1. Instantaneous Manpower Equation:

$$m(t) = 2Kat \cdot e^{-at^2} = \frac{K}{t_d^2} t \cdot e^{-t^2/(2t_d^2)} \ [\text{Persons}]$$

Where K is total lifecycle effort in Person-Years, t_d is development time (time of peak manpower), and $a = \frac{1}{2t_d^2}$.

2. Verified Peak Manning (m_0 at $t = t_d$):

$$m_0 = \frac{K}{t_d\sqrt{e}} \approx 0.6065 \frac{K}{t_d} \ [\text{Persons}]$$

: K.K. Aggarwal & Yogesh Singh (3rd ed.), §4.6.1, printed pp. 182–185; Companion Slides 123–135.

. Difficulty and Manpower Build-Up Metrics (D, D_0)

mic Definition & Technical Specification: The difficulty metric D is defined as total effort divided by the square velopment time; the distinct manpower build-up metric D_0 is total effort divided by the cube of development

ematical Formulation:

1. Difficulty Index:

$$D = \frac{K}{t_d^2} \, [\text{Persons/Year}]$$

2. Manpower Build-Up Metric:

$$D_0 = \frac{K}{t_d^3} \, [\text{Persons/Year}^2]$$

3. Empirical Values shown in the course:

$D_0 = 8$: new software with many interfaces/interactions.

$D_0 = 15$: new standalone software.

$D_0 = 27$: rebuilding existing software.

: K.K. Aggarwal & Yogesh Singh (3rd ed.), §4.6.2, printed pp. 185–187; Companion Slides 136–145.

Productivity Versus Difficulty

mic Definition & Technical Specification: Productivity Versus Difficulty represents Putnam's empirical tion demonstrating that overall software productivity is inversely related to project difficulty, scaled by an izational technology factor.

ematical Formulation:

1. Putnam's Productivity Relation:

$$P = \frac{S}{K} = C_k \cdot D^{-2/3}$$

Where S is delivered source lines of code, K is total effort, C_k is the technology factor, and D is project difficulty.

2. Engineering Implication: At fixed size and technology, the Putnam equations relate higher difficulty D to lower modeled productivity P.

: K.K. Aggarwal & Yogesh Singh (3rd ed.), §4.6.3, printed pp. 186–187; Companion Slides 146–148, 174–179.

ملاحظة أكاديمية هامة (Academic Note)

الشرح المفاهيمي وفخ الامتحان:

- **مفهوم المصطلح:** في علاقة النموذج، ومع تثبيت عامل التقنية، ترتبط زيادة مقياس الصعوبة بانخفاض الإنتاجية المقدّرة.
- **نقطة تمييز:** الأس في العلاقة هو $-2/3$؛ لا تستنتج من المعادلة وحدها أسباباً سلوكية أو إدارية.

The Trade-off Between Time Versus Cost

mic Definition & Technical Specification: The Putnam time–effort relation gives a fourth-power trade-off: for size and technology factor, required effort varies inversely with the fourth power of development time.

ematical Formulation:

1. The Invariant Relationship: From Putnam's software equation $S = C_k K^{1/3} t_d^{4/3}$:

$$K \cdot t_d^4 = \left(\frac{S}{C_k}\right)^3 = \text{Constant}$$

2. The Fourth-Power Law of Schedule Compression:

$$K \propto \frac{1}{t_d^4}$$

Reducing development time to $0.80 t_d$ changes modeled effort by a factor of $\frac{1}{(0.80)^4} \approx 2.441$, an increase of about 144.1% under these fixed conditions.

: K.K. Aggarwal & Yogesh Singh (3rd ed.), §4.6.4, printed pp. 187–188; Companion Slides 149–150, 166–173.

Development Sub-Cycle

mic Definition & Technical Specification: The Development Sub-Cycle is the separate development curve used del effort through delivery; the source relates its total effort K_d to full-project effort by $K = 6K_d$. The full- t Rayleigh curve's cumulative effort at t_d is a different quantity, approximately $0.3935K$.

ematical Formulation:

1. Mathematical Integration:

$$E(t_d) = \int_0^{t_d} m(t)dt = K(1 - e^{-1/2}) \approx 0.3935K$$

2. Development Sub-Cycle Effort:

$$K = 6K_d \quad \Longrightarrow \quad K_d = K/6$$

3. Development-Curve Timing:

$$t_{od} = t_d/\sqrt{6}$$

The cumulative effort $E(t_d)$ on the full-project curve and the separately defined development-sub-cycle effort K_d are different quantities.

: K.K. Aggarwal & Yogesh Singh (3rd ed.), §4.6.5, printed pp. 188–191; Companion Slides 151–165.

📝 **ملاحظة أكاديمية هامة (Academic Note)**

الشرح المفاهيمي وفخ الامتحان:

- **مفهوم المصطلح:** التمييز بين مرحلة التطوير ومرحلة الصيانة في دورة حياة النظام؛ لا تساو بين الجهد التراكمي على منحنى ال حتى t_d ($\approx 0.3935K$) وبين جهد دورة التطوير $K_d = K/6$.
- **فخ الامتحان:** انتبه للتفريق في الرموز: K هو الجهد الكلي لدورة الحياة الكاملة، بينما K_d هو جهد مرحلة التطوير فقط. عند حس ذروة مرحلة التطوير، تأكد من استخدام K_d في المعادلة.

---

e 6: Software Risk Management

What is Risk?

mic Definition & Technical Specification: A software risk is a potential future problem involving uncertainty and le loss to project objectives such as cost, schedule, or technical performance.

ematical Formulation — Risk Exposure (RE):

1. Quantitative Risk Exposure:

$$RE = P \times L \quad (\text{or } RE = \text{Probability} \times \text{Impact})$$

Where P is probability (0.0 to 1.0) and L is financial or schedule loss.

2. Known vs. Predictable vs. Unpredictable Risks:

Known Risks: Uncovered after careful evaluation of project plan and constraints (e.g., unrealistic delivery date).

Predictable Risks: Extrapolated from past project history (e.g., staff turnover).

Unpredictable Risks: Events that cannot reasonably be identified in advance from available project information.

: K.K. Aggarwal & Yogesh Singh (3rd ed.), §4.7, printed p. 194; Companion Slides 180–183.

Typical Software Risks

mic Definition & Technical Specification: Typical Software Risks are recurring threats that may prevent a are project from meeting its objectives. The chapter groups examples into five categories.

ve categories and examples are:

Dependencies on outside agencies or factors: Availability of trained, experienced people; customer-furnished items or information.

Requirement issues: Lack of clear product vision; unprioritized or changing requirements.

Management issues: Inadequate planning; unclear project ownership; poor communication.

Lack of knowledge: Inadequate training or application-domain experience.

Other risk categories: Turnover of essential personnel; inadequate testing facilities; unachievable performance requirements.

: K.K. Aggarwal & Yogesh Singh (3rd ed.), §4.7.2, printed pp. 195–196; Companion Slides 184–189.

> 📝 **ملاحظة أكاديمية هامة (Academic Note)**
>
> **الشرح المفاهيمي وفخ الامتحان:**
>
> - **مفهوم المصطلح:** أمثلة على فئات مخاطر قد تظهر في مشاريع البرمجيات.
> - **فخ الامتحان:** هذه هي فئات الفصل الخمس؛ لا تستبدلها بقائمة مخاطر من تصنيف أو كتاب آخر.

Risk Management Activities

mic Definition & Technical Specification: The chapter groups risk-management activities into Risk Assessment isk Control. Assessment consists of Risk Identification, Risk Analysis, and Risk Prioritization; Control consists of Management Planning, Risk Resolution, and Risk Monitoring.

wo-Phase Framework:

1. Risk Assessment:

Risk Identification: Uncovering project hazards using checklists, interviews, and taxonomies.

Risk Analysis: Estimating probability (P) and loss (L).

Risk Prioritization: Ranking hazards via Risk Exposure ($RE = P \times L$).

2. Risk Control:

Risk Management Planning: Preparing a plan for dealing with each significant risk.

Risk Resolution: Executing the plan for dealing with a risk.

Risk Monitoring: Tracking risks during the project.

: K.K. Aggarwal & Yogesh Singh (3rd ed.), §4.7.3, printed pp. 196–198; Companion Slides 188–193.

exact difference between Physical LOC and Logical SLOC, citing why LOC is language-dependent.
wn the 5 information domain characteristics of Albrecht's FPA and their Average complexity weights from memory.
the leading coefficients and exponents of Basic COCOMO across Organic, Semidetached, and Embedded modes.
ow COCOMO II calculates its scaling exponent B from the 5 scale factors ($B = 0.91 + 0.01 \sum SF$).
the Rayleigh peak manpower equation $m_0 = \frac{K}{t_d \sqrt{e}}$ and state Putnam's fourth-power law of time-effort trade-off (
).
sk Exposure ($RE = P \times L$) and contrast Risk Assessment with Risk Control.